

QUANTAMENTALS

Combining Technical and Fundamental Analysis in a Quantitative Framework for Better Investment Results

by Christopher Cain, CMT and Laurence Connors | 2020 Charles H. Dow Award Winner



About the Author | Christopher Cain, CMT

As a Senior Quantitative Researcher at Connors Research, Chris develops and presents quantitative trading models for select institutional investors and high-net-worth individuals. He is an expert on trading system design and development and shares his insights weekly in the Connors Research Traders Journal.

Mr. Cain is the creator and lead instructor for Trading Markets Python “Programming For Traders” course, where he teaches fellow financial professionals how to use Python to improve their investment process. He’s also the lead instructor with Larry Connors of the course, “Quantamentals; The Next Forefront of Trading and Investing.”

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Mr. Cain has 10 years of institutional trading experience, serving as a market maker in a range of fixed income instruments at various maturities. He is passionate about quantitative trading, investing, data science, coding, and behavioral finance.



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Laurence Connors is Chairman of The Connors Group, and the Managing Partner of Connors Research LLC with special focus on financial markets research. His companies have twice been awarded by the Entrex Private Company Index for being among the 10 fastest-growing private companies.

Mr. Connors has over 35 years of experience working in the financial markets industry. He started his career in 1982 at Merrill Lynch as an Investment Advisor, and later moved on to become a Vice President with Donaldson, Lufkin, Jenrette (DLJ), where he worked with the Investment Services Group from October 1990 to March 1994.

Mr. Connors is widely regarded as one of the leading educators in the financial markets industry. He has authored over 20 books on market strategies and volatility trading, including *Short-Term Trading Strategies That Work* and *Street Smarts* (with Linda Raschke). *Street Smarts* was selected by Technical Analysis of Stocks and Commodities magazine as one of “The Classics” for trading books written in the 20th century.

His most recent book *The Alpha Formula: Beat the Market with Significantly Less Risk* (with Chris Cain, CMT) was published in 2019. *The Alpha Formula* is the precursor for the development of “Quantamentals” which systematically combines fundamental, technical, and quantitative analysis for investment portfolios. Quantamentals is the 2020 winner of the Charles H. Dow Award.

Mr. Connors has been featured and quoted in the Wall Street Journal, New York Times, Barron’s, Bloomberg TV & Radio, Bloomberg Magazine, Dow Jones Newswire, Yahoo Finance, E-Trade Financial Daily, Technical Analysis of Stocks and Commodities, and many others. Mr. Connors has also been a featured speaker at a number of major investment conferences over the past three decades.

Abstract

Fundamental and Technical analysis are often thought of as separate, competing ideologies. We regard this world view as short-sighted. Our research shows that combining fundamental and technical analysis in a quantitative, rules-based framework leads to greatly improved performance. Furthermore, combining known factors whether of the fundamental or technical variety offers significant benefits and leads to market-beating performance over the last 16+ years.

Introduction

Factor investing has taken the investment management industry by storm. In this paper, we show that traditional factor investing can be greatly enhanced by combining factors in an intelligent way. More specifically, combining fundamental factors with technical factors leads to large increases in performance.

These different disciplines, fundamental and technical analysis, have rarely been combined throughout history. Fundamental investors often view technical analysis with a large degree of skepticism. Technicians are often of the opinion that “price is the only thing that pays.” Technicians believe that making investment decisions based on market generated signals such as price trends, overbought/oversold oscillators, and volatility is the most efficient way to generate outperformance and manage risk.

We regard these rather narrow world views as incomplete. Our research shows there are large advantages to incorporating both technical and fundamental factors intelligently into an investment process. Combining these disciplines in a quantitative, rules-based way has synergistic effects and gives an investor the best of both worlds.

Part 1: Meet the Factors

Fundamental Factors

We define fundamental factors as the metrics found on a firm’s quarterly balance sheet, income statement or other financial reports.

Value

Value, perhaps the most famous factor, is the tendency for relatively cheap stocks to outperform relatively expensive stocks over time. This factor has been around for almost a century, beginning with the “father of value investing,” Columbia Professor Benjamin Graham. His groundbreaking works “Security Analysis” (1934) and “The Intelligent Investor” (1949) laid the groundwork for the value investing philosophy.

The traditional academic definition of the value factor is to sort stocks based on their book-to-market ratio. The Fama-French three-factor model was introduced in the 1992 paper “The Cross Section of Expected Stock Returns” by Eugene Fama and Kenneth French. The Fama-French three-factor model uses the value factor, along with the market and size factors, to describe stock returns.¹

Quality / Profitability

A more recent fundamental factor identified by academics and used by practitioners long before its publication is the profitability or quality factor. This is the observation that investing in highly profitable firms has led to significantly higher returns compared to firms of lower profitability. Common metrics to measure a firm’s profitability include gross profitability, return on equity and return on invested capital.

The quality factor takes this idea one step further and shows that not only does profitability drive excess returns but also other metrics of strong financial standing do as well. Common metrics to measure a firm’s financial standing include low debt-to-asset ratios, the stability of earnings and low accruals.

The seminal academic work on the profitability premium comes from Robert Novy-Marx and his 2013 paper, “The Other Side of Value: The Gross Profitability Premium.” His work looked at gross profits, defined as sales minus the cost of goods sold divided by current assets, over the period of 1962 to 2010. Novy-Marx found that the most profitable firms earned returns of 0.31% more per month compared to the least profitable firms.² Furthermore, he found that accounting for profitability dramatically increased the performance of value-based strategies, an insight we will utilize in the model we will build in this paper.

¹ Fama, Eugene, Kenneth French, 1992, The Cross Section of Expected Stock Returns, The Journal of Finance

² Novy-Marx, Robert, 2012, The Other Side of Value: The Gross Profitability Premium, Journal of Financial Economics

Technical Factors

We define technical factors as those that are derived from the market itself. These factors either use price directly or a derivative of price, such as volatility.

Cross-Sectional Momentum

Cross-sectional momentum is the tendency for assets that have had the strongest performance in the recent past to continue to outperform going forward. On the other hand, assets that have had the weakest performance over the recent past tend to continue to underperform. This type of momentum, sometimes called relative strength momentum or simply relative strength, compares the performance of an asset to a larger universe of assets. This is slightly different than time-series momentum, which looks at an asset's performance compared to its own past history.

The seminal study on the momentum effect was conducted by Narasimhan Jegadeesh and Sheridan Titman in their 1993 paper – “Returns of Buying Winners and Selling Losers: Implications for Stock Market Efficiency.” This paper formed long/short portfolios based on past performance (momentum), showing large outperformance that was unable to be explained by existing academic models.³

The traditional academic definition of momentum, especially as it is applied to individual equities, measures the total returns over the last 12 months, skipping the most recent month. Skipping the most recent month is to account for the well-known mean reversion effect over shorter time frames.

Low Volatility / Low Beta

The low volatility factor is the empirical observation that “defensive” stocks (low-volatility, low-risk) have delivered both higher returns and higher risk-adjusted returns compared to “aggressive” stocks (high-volatility, high-risk).

This is a large blow to the original academic pricing model – the capital asset pricing model (CAPM). CAPM states that there is a positive relationship between risk and return, as a “rational” investor should demand a higher return to compensate for accepting more risk. Not only does this fail to hold up in the real world, but the empirical results are the opposite. Low volatility (risk) stocks tend to lead to higher returns. The outperformance is even more dramatic when returns are viewed on a risk-adjusted basis.

³ Jegadeesh, Narasimhan, Sheridan Titman, 1993, Returns of Buying Winners and Selling Losers: Implications for Stock Market Efficiency, The Journal of Finance

Academic evidence for the low volatility premium includes the 2016 study “Understanding Defensive Equity” written by Robert Novy-Marx. Novy-Marx ranked stocks by quintiles of either volatility or market beta, and showed that the most volatile/highest beta quintile dramatically underperformed the rest of the stocks.⁴

Further evidence was provided by Andrea Frazzini and Lasse Heje Pedersen in their 2014 paper “Betting Against Beta.” The professors formed portfolios that went long low beta stocks (leveraged to a beta of 1) and shorted high-beta stocks (deleveraged to a beta of 1). This market neutral portfolio realized a Sharpe ratio of 0.78 from 1926 to 2012. Frazzini and Pedersen then expanded this research to not only include the US but investigated 10 international equity markets and found similar results.⁵

Time-Series Momentum

Time-series momentum is using a security’s own past performance to dictate long or short positions. An example time-series momentum rule would be to go long if the security is trending higher and getting out, or going short, if the security is trending lower. Time-series momentum is also called absolute momentum or simply trend following.

Many technical analysis techniques can be utilized to measure time-series momentum. Some examples include a simple rate of change, the security’s price compared to its moving average, dual moving averages, or the slope of a linear regression line, to name a few.

Time-series momentum is perhaps the oldest and most successful investment style in the hedge fund industry. Trend following is synonymous with the CTA industry, a category of hedge funds. These firms typically utilize trend following on a diversified group of futures markets to deliver uncorrelated returns. The performance of these hedge funds is particularly strong in crisis periods such as 2008. As such, they are useful as part of a large institutional investor’s portfolios.

Academics were a few decades behind practitioners in studying the effects of time-series momentum. In recent years, however, time-series momentum has been widely studied and proved to be a robust and statistically significant driver of performance. One recent study is “A Century of Evidence on Trend-Following Investing” by Brian Hurst, Yao Hua Ooi and Lasse H. Pedersen. In the paper, portfolios

⁴ Novy-Marx, Robert, 2014, Understanding Defensive Equity, NBER Working Papers

⁵ Frazzini Andrea, Lasse Heje Pedersen, 2014, Betting Against Beta, *Journal of Financial Economics*

were constructed by simply taking equally weighted one month, three months and twelve month total returns. The model went long assets that have shown positive recent performance and shorted assets that have shown negative recent performance.

This simple strategy was applied to 67 markets across four major asset classes (29 commodities, 11 equity indices, 15 bond markets, and 12 currency pairs). The time period for this study was an amazing 1880 to 2013. The results were nothing short of spectacular, showing remarkable consistency throughout the decades and delivering an annualized return of 14.9% with 9.7% volatility!⁶ Furthermore, returns were positive every decade and showed virtually no correlation to traditional equity or fixed income markets.

Part 2: Performance of Single Factors

We have established the factors we will be utilizing in our model, along with some quantitative evidence provided by academia regarding their effectiveness. We will now investigate some of the factors individually. We will then go on to show that combining the fundamental factors with the technical factors in an intelligent and quantitative, rules-based way leads to greatly enhanced performance.

Data and Investment Universe

All historical tests will cover the time period of January 2003 to September 2019. Data and analytics are provided by Quantopian.com. The universe for all tests run is Quantopian's "Q500US" universe, which contains the 500 most liquid US stocks based on trailing 200-day average dollar volume. This universe is reconstituted each month, avoiding survivorship bias.

Quality/Profitability Factors

We will begin by inspecting the results of buying stocks that have high quality/strong profitability. The metrics we will utilize are sourced from academic papers or books.

As mentioned previously, the seminal research on the quality factor was conducted by Robert Novy-Marx in his 2013 paper "The Other Side of Value: The Gross Profitability Premium." In the paper, the author used gross profitability, as defined by revenues minus cost of goods sold divided by assets. We use this as a quality/profitability metric.

⁶Hurst, Brian, Yao Hua Ooi and Lasse H. Pedersen, 2014 , A Century of Evidence on Trend-Following Investing, The Journal of Portfolio Management

Other common quality/profitability metrics involve how efficiently a firm utilizes its capital. A metric used by the highly successful hedge fund manager Joel Greenblatt in his book “The Little Book That Beats The Market” was return on invested capital (ROIC).⁷ We will investigate this measure as well. Similar to ROIC, we will also include the popular metric return on equity (ROE), another measure of how efficiently a firm utilizes its capital.

For these tests, we will create two long-only portfolios, one that buys the top 50 stocks based on our three quality metrics and one that buys the bottom 50. These portfolios will be rebalanced once a month.

Table 1: Quality Factor Test

Start Date	01/01/2003					
End Date	09/30/2019					
	ROE		ROIC		Gross Profitability	
	Top 50	Bottom 50	Top 50	Bottom 50	Top 50	Bottom 50
Annual Return	11.60%	6.70%	12.40%	5.90%	10.10%	6.10%
Cumulative Returns	521.90%	197.00%	608.70%	158.80%	401.90%	171.10%
Annual Volatility	18.40%	26.80%	20.10%	26.50%	19.60%	22.20%
Sharpe Ratio	0.69	0.38	0.68	0.35	0.59	0.38
Max Drawdown	-55.70%	-70.40%	-58.60%	-69.30%	-53.70%	-60.40%

As you can see, separating firms by profitability/quality metrics produces large spreads between the highest and lowest quality firms. Furthermore, higher-quality firms display markedly lower volatility, which when combined with higher returns results in significantly higher Sharpe ratios.

Value Factors

Next, we will inspect the performance of some stand-alone value factors. For our value metrics, we will utilize a couple of measures that incorporate both top line and bottom line statistics.

We start with EBIT (earnings before interest and taxes) to Enterprise Value (the value that a private investor would be forced to pay to buy a company, including equity and debt). This measure was again popularized by Joel Greenblatt, and this is the other metric utilized in his work “The Little Book That Beats The Market.”

⁷ Greenblatt, Joel, 2005, The Little Book That Beats The Market, Wiley

Next, we will include a top-line measure for value: price to sales. This metric was popularized by Jim O'Shaughnessy in his book "What Works on Wall Street" amongst others.⁸ Price-to-sales is calculated by taking a company's market capitalization and dividing it by the company's total sales or revenues over the past twelve months.

Notice we include both top-line (sales/revenues) and bottom line (earnings before interest and taxes) statistics to measure value. We again create long-only portfolios, separate firms into the 50 cheapest and 50 most expensive by these metrics. Our portfolios are rebalanced monthly.

Table 2: Value Factor Tests

Start Date	01/01/2003			
End Date	09/30/2019			
	EBIT/EV		Price/Sales	
	50 Cheapest	50 Most Expensive	50 Cheapest	50 Most Expensive
Annual Return	10.50%	6.90%	12.40%	9.70%
Cumulative Return	430.70%	204.70%	602.60%	368.20%
Annual Volatility	23.00%	25.80%	20.00%	30.80%
Sharpe Ratio	0.55	0.39	0.68	0.45
Max Drawdown	-67.00%	-66.30%	-60.20%	-68.90%

While not as dramatic as the quality factor for the last 16+ years, you can see the cheapest firms outperformed the most expensive firms based on our value metrics.

Cross-Sectional Momentum

We now move on to test cross-sectional momentum. The traditional academic definition of cross-sectional momentum measures the total return of a stock over the last 12-months and excludes the most recent month.

The exclusion of the most recent month in academic research is to account for the well-known tendency of stocks to mean revert over this shorter-term time frame. To remain consistent, we will skip the most recent month as well.

⁸ O'Shaughnessy, Jim, 1997, What Works on Wall Street, McGraw-Hill

For our tests, we will inspect both 12-month and 6-month momentum lookbacks. We again form long-only portfolios, investing in the 50 firms with the highest and lowest momentum readings with a monthly rebalance.

Table 3: Momentum Factor Tests

Start Date	01/01/2003			
End Date	09/30/2019			
	Momentum - 6 Month, 1 Month Skip		Momentum - 12 Month, 1 Month Skip	
	50 High Momentum	50 Low Momentum	50 High Momentum	50 Low Momentum
Annual Return	12.00%	4.90%	7.70%	4.50%
Cumulative Return	567.60%	121.70%	247.40%	109.40%
Annual Volatility	24.30%	27.50%	27.00%	27.00%
Sharpe Ratio	0.59	0.31	0.41	0.3
Max Drawdown	-62.40%	-68.40%	-70.00%	-71.40%

Consistent with academic research, we see stocks with strong recent performance outperforming stocks with weak recent performance over our time period.

Low Volatility

To inspect the low volatility factor, we form long-only portfolios of the 50 stocks with the highest and lowest historical volatility with a monthly rebalance. We utilize two lookback windows, 100 days and 200 days, for our volatility calculations.

Table 4: Volatility Factor Tests

Start Date	01/01/2003			
End Date	09/30/2019			
	Volatility - 100 Day Lookback		Volatility - 200 Day Lookback	
	50 Low Volatility	50 High Volatility	50 Low Volatility	50 High Volatility
Annual Return	9.70%	5.50%	9.90%	3.10%
Cumulative Return	369.30%	142.80%	383.20%	67.30%
Annual Volatility	11.50%	32.10%	11.40%	33.90%
Sharpe Ratio	0.86	0.33	0.89	0.26
Max Drawdown	-40.40%	-68.50%	-37.10%	-78.70%

The spread between the least and most volatile stocks is dramatic in both the 100-day and 200-day lookbacks. Not only do the lower volatility stocks produce significantly higher returns, but they do so with less volatility and drawdown, leading to a significant jump in Sharpe ratios.

Time-Series Momentum

To inspect time-series momentum, we will apply two simple time-series momentum rules to the overall US stock market, represented by the ETF "SPY." We will utilize both raw total return momentum (rate of change) signals as well as price vs. moving average signals.

In our simple moving average tests, we will go long SPY if its price is above its moving average and switch to SHY (1-3yr US Treasury bonds) if its price is below its moving average. For our total return momentum (ROC) tests, we will go long SPY if its total return over the lookback period is positive and switch to SHY if the total return over the lookback period is negative. This signal is checked once a month, at the end of the month.

We will inspect the use of both 100-day and 200-day moving averages and 6-month and 12-month momentum.

Table 5: Time Series Momentum Tests

Start Date	01/01/2003				
End Date	09/30/2019	Moving Averages		Momentum (ROC)	
	SPY	100-day MA	200-day MA	6-Month	12-Month
Annual Return	9.60%	8.50%	10.20%	9.60%	8.90%
Cumulative Return	365.40%	288.30%	405.40%	361.40%	313.90%
Annual Volatility	17.80%	11.20%	12.00%	11.90%	12.90%
Sharpe Ratio	0.61	0.78	0.86	0.83	0.72
Max Drawdown	-55.20%	-20.40%	-20.80%	-19.70%	-19.60%

Moving Average Rule: Is Price above the Moving Average

Momentum Rule: Is Total Return Positive

Risk Off Asset: SHY (1-3yr US Treasuries)

Results here are typical of applying time-series momentum rules, most notably the addition of these rules allows us to sidestep large bear markets. Consistent with our prior research, applying time-series momentum results in marked decreases in volatility and max drawdown compared to buy and hold alone.⁹

Summary of Individual Factor Tests

In this section, we tested both Fundamental and Technical factors in isolation. We witnessed the anticipated results with high quality beating low quality, cheap beating expensive, high momentum beating low momentum and low volatility beating high volatility. We also observed the risk-reducing nature of applying simple time-series momentum rules.

Part 3: Combining the Factors and Building Our Model

We will now move on to the heart of the paper, the fact that the combination of fundamental and technical analysis in a quantitative, rules-based way leads to dramatic performance increases.

We will begin by combining the two fundamental factors – quality and value. We will then move on to combine these fundamental factors with technical factors: low volatility, time-series momentum and finally cross-sectional momentum.

⁹ Cain, Christopher and Larry Connors, 2019, The Alpha Formula, TradingMarkets Publishing

Combining the Fundamental Factors - Quality and Value

Research has shown that combining quality and value has synergistic effects. Robert Nozy-Marx, in his aforementioned paper “The Other Side of Value: The Gross Profitability Premium” showed that this combination resulted in increased performance. The combination of quality and value also allows an investor to avoid the so-called “value trap,” firms that look cheap but have little chance of a turnaround.

The combination of value and quality also explains much of Warren Buffett’s spectacular success throughout the years. After all, Mr. Buffett’s philosophy isn’t simply to buy cheap stocks as some naively believe. He instead opts for quality companies that are trading at relatively cheap valuations.

Finally, Joel Greenblatt also combines quality and value metrics in his work, opting for companies that have high ROIC and high EBIT/EV ratios. Mr. Greenblatt is essentially sorting for companies that are both highly profitable and trading at relatively cheap valuations.

For the rest of this paper, we will stick with ROIC to represent quality and EBIT/EV to represent value, just as Mr. Greenblatt did. This combination has been in the public domain for many years.

In the first step in building our Quantamental model, we combine the value (EBIT/EV) and quality (ROIC) metrics. We first rank our 500 stocks by the value metric, with 500 being the cheapest and 1 being the most expensive. We then rank our 500 stocks by the quality metric, with 500 being the firm with the highest quality and 1 being the firm with the lowest quality. We then simply sum the rankings, rebalancing into the 50 firms that have the highest combined rank: firms with both high quality and low valuations. To stay consistent with the tests previously run, we will rebalance our portfolio monthly.

Table 6: Quality + Value

Combination of Quality and Value	
Start Date	01/01/2003
End Date	09/30/2019
Quality	ROIC
Value	EBIT/EV
Annual Return	11.30%
Cumulative Returns	498.80%
Annual Volatility	21.90%
Sharpe Ratio	0.60
Max Drawdown	-64.60%

This combination results in annual returns of 11.3%, volatility of 21.9%, a Sharpe ratio of 0.60 and a significant drawdown of -64.60%. We will now demonstrate how adding technical factors greatly improves these results.

Performance of Combining Value, Quality, and Low Volatility

As we already witnessed, simply buying stocks with the lowest historical volatility has been a great strategy over the last 16 years. These stocks have delivered higher absolute returns and much higher risk-adjusted returns compared to the overall market. We will utilize this insight in our evolving model. In addition to the quality and value metrics, we will now add low volatility.

The methodology here is the same. We will rank each stock by quality and value, the same as before, but this time add in a ranking for low volatility. We will apply a 100-day lookback to calculate volatility, sticking with the standard deviation of daily percent returns over this time frame. The stock with rank 500 will be the lowest volatility stock and the stock with rank 1 will be the highest volatility stock

We will then simply sum our three rankings – quality, value, and low volatility, and invest in the top 50 stocks with the highest combination rank. We again rebalance the portfolio monthly.

Table 7: Quality, Value and Low Volatility

Combination of Quality, Value and Low Vol	
Start Date	01/01/2003
End Date	09/30/2019
Quality	ROIC
Value	EBIT/EV
Low Volatility	100-Day Lookback
Annual Return	11.90%
Cumulative Returns	557.50%
Annual Volatility	14.50%
Sharpe Ratio	0.85
Max Drawdown	-40.30%

We witness a significant increase in performance by adding the low volatility factor. Annualized returns went from 11.3% to 11.9%, volatility went from 21.9% to 14.5% and max drawdown went from -64.6% to -40.3%. This resulted in a large jump in the Sharpe ratio, going from 0.60 to 0.85.

Performance of Combining Value, Quality, Low Volatility, and Time-Series Momentum

The next factor we will add to our model is time-series momentum or trend following. We will do this in the form of a trend following “regime filter.” This well-known technique will first check if the overall market is trending higher. Only then will the model take new entries in our monthly rebalance. If the overall market is trending lower, no new entries are taken.

We now have to introduce a “risk-off” asset, something to rotate into when the trend of the overall market is down and we aren’t buying more stocks. For this purpose, we will simply use SHY (1-3yr US Treasuries). Other “risk-off” assets such as longer duration US Treasuries or US aggregate bonds can also be utilized and will help historical test results.

There are many ways we can measure if the overall market is trending higher or lower. Instead of looking for the optimal way to measure this, we will instead use a simple moving average with a 100-day lookback. We will use the ETF “SPY” to represent the market.

If the price of SPY is above its 100-day moving average, we will conclude that the market is trending higher and our model will take new entries. If the price of SPY is below its 100-day moving average, we will conclude that the trend of the market is lower and our model *will not take new entries*.

A note on the logic here – if SPY is below its 100-day moving average, we sell any stocks that fell out of our top 50 based on our value, quality and low volatility factors. These stocks are not replaced, with that capital instead allocated to SHY. If SPY is below its 100-day moving average and the stock remains in the top 50, it is held.

Table 8. Quality, Value, Low Volatility and Time-Series Momentum

Combination of Quality, Value, Low Vol and Time-Series Momentum	
Start Date	01/01/2003
End Date	09/30/2019
Quality	ROIC
Value	EBIT/EV
Low Volatility	100-Day Lookback
Time-Series Momentum	100-Day Moving Average
Annual Return	10.70%
Cumulative Returns	445.30%
Annual Volatility	10.90%
Sharpe Ratio	0.99
Max Drawdown	-25.60%

Adding time-series momentum has the anticipated effect, significantly lowering the volatility and drawdown. The volatility now decreased from 14.5% to 10.9% and the max drawdown went from -40.3% to -25.60%. Our Sharpe ratio is now 0.99.

Adding Cross-Sectional Momentum: Double Sort

The last step in our model is to incorporate cross-sectional momentum. We will utilize this in a “double sort.”

We first rank our stocks by our quality, value and low volatility factors, taking the top 50 stocks with the highest combined metrics. We will then rank those top 50 stocks using cross-sectional momentum. Our momentum measure will be the trailing 6-month total returns, skipping the last month. We will then take the top 20 stocks with the highest momentum scores. We are left with 20 stocks, which will be our final portfolio. We continue applying the time-series momentum rule, only taking new entries if the price of SPY is above its 100-day moving average.

Table 9: Quality, Value, Low Volatility, Time-Series Momentum and Cross-Sectional Momentum

Combination of Quality, Value, Low Vol, Time-Series Momentum and Cross-Sectional Momentum	
Start Date	01/01/2003
End Date	09/30/2019
Quality	ROIC
Value	EBIT/EV
Low Volatility	100-Day Lookback
Time-Series Momentum	100-Day Moving Average
Cross-Sectional Momentum	6-Month Lookback 1-Month Skip
Annual Return	13.30%
Cumulative Returns	701.40%
Annual Volatility	12.00%
Sharpe Ratio	1.10
Max Drawdown	-23.00%

Our final rule, adding cross-sectional momentum, increases the returns by 2.6% per year while decreasing the max drawdown from -25.6% to -23.0%. Our Sharpe ratio is now 1.10 over the last 16+ years.

Improvement Every Step of the Way

Notice how the model results improved with each factor we added. We started with fundamental factors only, which were quality and value. We then added a technical factor: low volatility. We then moved on to add a time-series momentum rule in the form of a trend-following regime filter. This had the desired effect: a reduction of risk and max drawdowns. Finally, we added a cross-sectional momentum rule via a double sort, leading to higher returns.

Our Sharpe Ratio nearly doubled by combining fundamental and technical factors in a thoughtful way, going from 0.60 for the fundamental factors only to 1.10 in our complete model. The following table displays model results for every new factor added.

Table 10: Incremental Improvements

Start Date	01/01/2003			
End Date	09/30/2019			
Quality	ROIC	ROIC	ROIC	ROIC
Value	EBIT/EV	EBIT/EV	EBIT/EV	EBIT/EV
Low Volatility	N/A	100-Day Lookback	100-Day Lookback	100-Day Lookback
Time-Series Momentum	N/A	N/A	100-Day MA	100-DAY MA
Cross-Sectional Momentum	N/A	N/A	N/A	6-mo 1-mo Skip
Annual return	11.30%	11.90%	10.70%	13.30%
Cumulative returns	498.80%	557.50%	445.30%	701.40%
Annual volatility	21.90%	14.50%	10.90%	12.00%
Sharpe ratio	0.60	0.85	0.99	1.10
Max drawdown	-64.60%	-40.30%	-25.60%	-23.00%

Our Complete Quantamental Model

Going step by step and combining known factors, we built a complete trading model incorporating both fundamental and technical factors. Our model utilizes quality, value, low volatility, time-series momentum and cross-sectional momentum in a complete trading strategy. Furthermore, we observed that combining the fundamental and technical factors leads to greatly improved performance, nearly doubling our Sharpe Ratio.

Summary of the rules:

- 1) We start with a universe of the 500 most liquid US Stocks. This is Quantopian's "Q500US" universe. This universe is derived by taking the 500 US stocks with the largest average 200-day dollar volume, reconstituted monthly, capped at 30% of the equities derived from one sector.
- 2) We then rank our stocks 1-500, based on the quality, value and low volatility factors. The stock ranked 500 would be the most attractive of these attributes and the stock ranked 1 would be the least attractive.
 - a. Quality - Rank stocks by ROIC, the higher the better
 - b. Value - Rank stocks by EBIT/EV, the higher the better
 - c. Volatility - Rank stocks by trailing 100-day standard deviation, the lower the better

- 3) Add up the three rankings and take the top decile. We are now left with 50 stocks that have a combination of high quality, low valuation, and low volatility.
- 4) Of our 50 stocks, take the top 20 based on cross-sectional momentum. This is measured by stocks with the highest 6-month total return, skipping the last month.
- 5) Every month we rebalance our portfolio, selling any stock we currently hold that didn't make the top 20 list based on the logic above and buying stocks that have since made the list. Stocks are equally weighted.
- 6) We only take new entries if our time-series momentum regime filter is passed. For our time-series momentum regime filter, we simply use SPY's price compared to its 100-day moving average. If the price of SPY is above its 100-day moving average, we take new entries. If the price of SPY is below its 100-day moving average, no new entries are taken.
- 7) Any capital not allocated to stocks gets allocated to SHY (1-3yr US Treasuries).
- 8) Assumptions include a beginning portfolio balance of \$1,000,000 and commissions of \$0.005 per share with a minimum trade ticket cost of \$1. This models the real-life commission schedule of Interactive Brokers.

Performance

Table 11: Performance of Quantamentals Model vs SPY, 2003-2019

Start Date	1/1/2003	
End Date	9/20/2019	
Summary Statistics		
	SPY	Quantamentals Model
Annualized Return	9.60%	13.30%
Cumulative Returns	365.40%	701.40%
Annualized Volatility	17.80%	12.00%
Sharpe Ratio	0.61	1.10
Calmar Ratio	0.17	0.58
Max Drawdown	-55.20%	-23.00%
Sortino Ratio	0.85	1.55
Skew	-0.08	-0.5
Kurtosis	11.2	3.11
Daily Value at Risk	-2.20%	-1.50%
Annualized Alpha	0.0%	9.2%
Beta	1.0	0.43

Chart 1: Quantamental Model vs SPY, Cumulative Equity Growth, 01/2003-09/2019

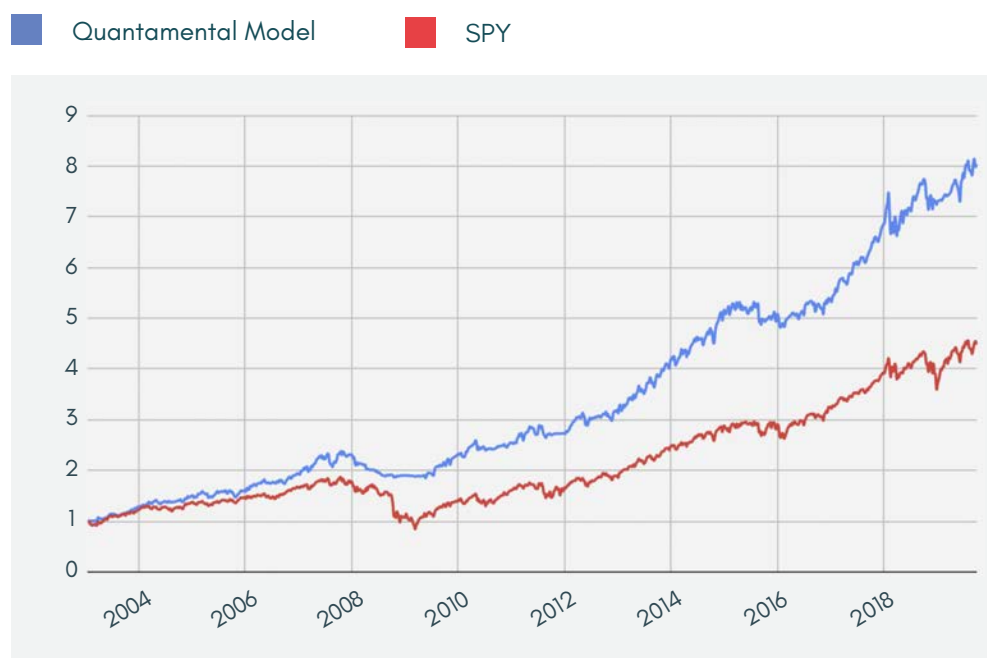


Table 12: Quantamental Model Monthly/Yearly Total Returns

	Jan	Feb	March	April	May	June	July	Aug	Sept	Oct	Nov	Dec	Total
2003	0.30	0.59	3.00	0.78	4.67	3.95	-3.62	3.20	-0.24	7.19	3.32	3.64	29.76
2004	0.82	3.70	1.36	0.56	-1.43	1.85	-0.30	0.06	1.65	-1.12	5.86	2.11	15.94
2005	-1.18	4.21	-0.56	-4.53	2.49	3.61	1.81	-0.14	-1.30	-3.58	4.38	0.53	5.42
2006	6.86	0.86	2.31	2.17	-1.77	0.77	0.20	1.37	-0.88	4.93	1.98	1.52	21.89
2007	5.08	-1.50	3.10	5.08	5.31	-1.90	-3.51	0.32	6.86	2.77	-3.63	-0.20	18.45
2008	-6.17	-1.00	-4.47	-1.17	-0.28	-1.42	-3.45	0.13	0.13	-1.23	0.49	-0.07	-17.21
2009	-0.10	-0.70	0.69	-0.43	2.87	-0.09	6.99	0.21	3.23	-1.46	6.09	2.88	21.67
2010	-1.50	4.30	4.65	2.64	-4.03	-1.78	1.08	-0.47	2.16	0.96	-2.61	4.91	10.27
2011	0.96	5.36	0.11	5.51	-0.08	-1.20	-1.58	-1.71	-1.16	0.95	0.06	-0.15	6.96
2012	3.54	5.29	2.70	2.16	-4.95	2.46	0.39	0.10	1.97	-1.89	3.72	0.87	17.18
2013	1.98	2.12	3.75	0.85	2.27	-0.08	5.77	-3.12	5.82	2.46	3.67	3.47	32.71
2014	-2.83	4.30	2.16	-0.11	3.91	0.21	-1.49	4.73	0.83	2.29	5.08	-0.15	20.26
2015	-0.34	3.79	-0.56	-2.91	0.80	-0.21	3.22	-6.74	-0.16	2.49	0.23	-0.45	-1.27
2016	-2.48	1.49	2.18	-0.21	-0.16	3.25	1.69	-0.93	0.34	-1.64	2.20	0.16	5.87
2017	3.10	4.44	-0.02	2.11	2.75	0.69	2.46	0.36	2.90	2.00	2.78	2.44	29.25
2018	6.07	-7.61	0.96	3.35	-0.11	1.27	2.31	5.26	1.00	-6.25	1.19	-0.33	6.37
2019	0.68	0.77	0.97	3.62	-5.90	6.33	2.22	1.27	-0.38				9.53

Robustness Check

For a robustness check, we will look at different variations of our model using various techniques for our time-series momentum regime filter. We will also inspect various lookback periods for our cross-sectional momentum sort.

Find four versions of our model in the following table using various techniques for our time-series momentum regime filter. Holding all other parameters constant, we inspect the results using a 100-day moving average, a 200-day moving average, 6-month total return momentum (ROC) and 12-month total return momentum (ROC).

Table 13: Time-Series Momentum Robustness Tests

Start Date	01/01/2003			
End Date	09/30/2019			
Quality	ROIC			
Value	EBIT/EV			
Low Volatility	100-Day Lookback			
Cross-Sectional Momentum	6-Month Lookback 1-Month Skip			
Time-Series Momentum	100-Day MA	200-Day MA	6-Month ROC	12-Month ROC
Annual return	13.30%	13.30%	14.70%	14.20%
Cumulative returns	701.40%	706.10%	891.30%	821.30%
Annual volatility	12.00%	12.60%	12.00%	12.70%
Sharpe ratio	1.10	1.06	1.20	1.11
Max drawdown	-23.00%	-21.10%	-16.20%	-16.30%

The performance of our model holds up against different variations of our time-series momentum filter as can clearly be seen. We actually see a bump in performance by using a 6-month total return momentum filter instead of a moving average.

Find four versions in the following table using various lookbacks for our cross-sectional momentum sort. Holding all other parameters constant, we inspect the results using a 3-month, 6-month, 9-month, and 12-month cross-sectional momentum lookback. To remain consistent, the most recent month is skipped in all of these tests.

Figure 15: Cross-Sectional Momentum Robustness Tests

Start Date	01/01/2003			
End Date	09/30/2019			
Quality	ROIC			
Value	EBIT/EV			
Low Volatility	100-Day Lookback			
Time-Series Momentum	100-Day Moving Average			
Cross-Sectional Momentum	3-Month	6-Month	9-Month	12-Month
Annual return	11.90%	13.30%	14.50%	13.40%
Cumulative returns	558.90%	701.40%	855.50%	716.40%
Annual volatility	11.70%	12.00%	12.10%	12.30%
Sharpe ratio	1.02	1.10	1.17	1.08
Max drawdown	-23.20%	-23.00%	-19.90%	-23.70%

Our model remains robust to different cross-sectional momentum lookbacks as well, with the 9-month lookback producing the best results.

Conclusion

In this paper, we first reviewed known drivers of unexplained returns, commonly known as factors: quality, value, low volatility, cross-sectional momentum, and time-series momentum. We reviewed the academic literature around these factors and ran tests examining the performance of these factors on a stand-alone basis. While each stand-alone factor delivered the expected result, the performance often came with high risks, steep drawdowns, and other drawbacks.

We then combined these factors in a thoughtful manner, showing improved performance every step of the way. We especially observed a bump in performance when the fundamental factors (quality and value) were combined with the technical factors (low volatility, time-series momentum and cross-sectional momentum).

We built our complete Quantamentals model from the ground up. We observed that the combination of fundamental and technical analysis in a quantitative, rules-based way leads to significant outperformance. The end result is a robust model that greatly exceeds the performance of the market benchmark.

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